



Ekman transport 60, 65  
El Niño effect 65  
electricity 119, 230–231  
erosion 44, 45  
estuaries 169, 172, 173, 176  
exoskeletons 126, 127  
exploration 33, 218–219, 224

## F

fat 209  
filter feeders 77, 79, 80, 88–89, 111, 120, 138, 173  
fins 82, 83, 89, 118, 119, 199  
fish 63, 64, 70, 71, 75, 78, 79, 80–89, 131, 135, 155, 156, 157, 160, 175, 177, 178, 179, 193  
    coral reefs 136–137, 138, 139  
    dark zone 100–101  
    on the ocean floor 102–103, 118–119  
    as prey 82–85, 95, 96, 97, 139, 167, 177, 184  
    twilight zone 98–99  
fish farms 231  
fishing 109, 232–235, 242, 243  
    overfishing 83, 109, 238  
fjords 47  
flamingoes 177  
flatfish 119, 161  
floats 112, 114  
floor of the oceans 24, 25, 28–29, 30, 36–37, 45  
    life on 102–103, 110–119  
food chains 78–79  
food webs 79  
fossils 46, 49  
    fossil fuels 230  
fracture zones 16, 17  
frigatebirds 97, 167  
fur 117, 209, 211

## G

Galápagos Islands 41, 155, 185, 219  
gannets 96, 166  
gases 25, 105, 173, 175  
    as fuel 230  
gastropods 120, 121  
*Geldermalsen* 227  
gill rakers 80, 88  
gills 80, 82, 88, 104, 105, 170  
    parasites 137  
glaciers 18, 47, 202–203, 204  
glass-making 229  
global conveyor 66, 67  
global warming 67, 240  
glow-in-the-dark animals 89, 98, 100, 130  
gold 227  
granite 24  
gravel 229  
gravity 146, 147  
Great Barrier Reef 134–135  
great white sharks 86–87  
guano 167  
guillemots 97, 166

Gulf of Mexico 183, 239  
Gulf Stream 60, 61, 62, 67  
gurnards 118  
gyres 52, 60

## H

hatchetfish 99  
Hawaii 26, 40, 42, 59, 221  
He, Zheng 218  
headlands 150, 151, 158  
heart urchins 160  
heat 50–51  
herrings 80, 81, 83, 109  
holdfasts 114  
hotspots 12, 40–41, 42  
human activity 158  
Humboldt Current 61  
hurricanes 55, 57, 174, 220, 241  
hydrogen 48

## I

ice 10, 18, 38, 46, 47, 49, 51, 66, 188, 189, 190–191, 194, 210  
ice fish 193  
icebergs 18, 202, 204–205, 207, 227  
icebreakers 191  
Iceland 40  
igloos 214  
Indian Ocean 9, 14–15, 132  
    animals 122, 136, 137, 139, 162, 168, 171, 183, 184, 185  
    currents 67  
    exploration 218  
    islands 40  
    trenches 29  
    tsunamis 38  
ink 123  
Inuit people 214–215  
invertebrates 120–121, 138–139, 192  
island arcs 34  
islands 10, 12, 15, 16, 17, 51, 142–143, 151, 171, 185, 200  
    and birds 166, 167  
    creation of 26, 34–35, 40, 41, 142–143, 151  
    and tsunamis 39

## J

Japan 29, 35, 39  
Java Trench 15  
jellyfish 76, 98, 130–131  
    as prey 168, 169

## K

kayaks 215  
kelp 114–115, 117  
krill 74, 75, 79, 91, 97, 192, 194  
Kuril Trench 33

## L

lagoons 134, 142–143, 159, 163, 169, 176  
landslides 38  
lantern fish 98  
larvae 75, 126  
lava 26, 30, 40  
levees 174  
lichens 155  
limestone 46, 132, 135  
limpets 78, 113, 154  
lionfish 137  
lizards 155, 184, 185  
lobsters 110, 126, 157  
longshore drift 158, 159

## M

mackerel 80, 85  
Maelstrom of Saltstraumen 147  
Magellan, Ferdinand 219  
mammals 94  
    *see also* Arctic fox, dolphins, dugong, manatee, polar bears, sea lions, sea otters, seals, walruses, whales  
manatee 183  
manganese 229  
mangroves 178–179, 239  
manta rays 79, 89  
mantle 24, 34, 36  
maps 29  
Mariana Trench 32–33, 225

